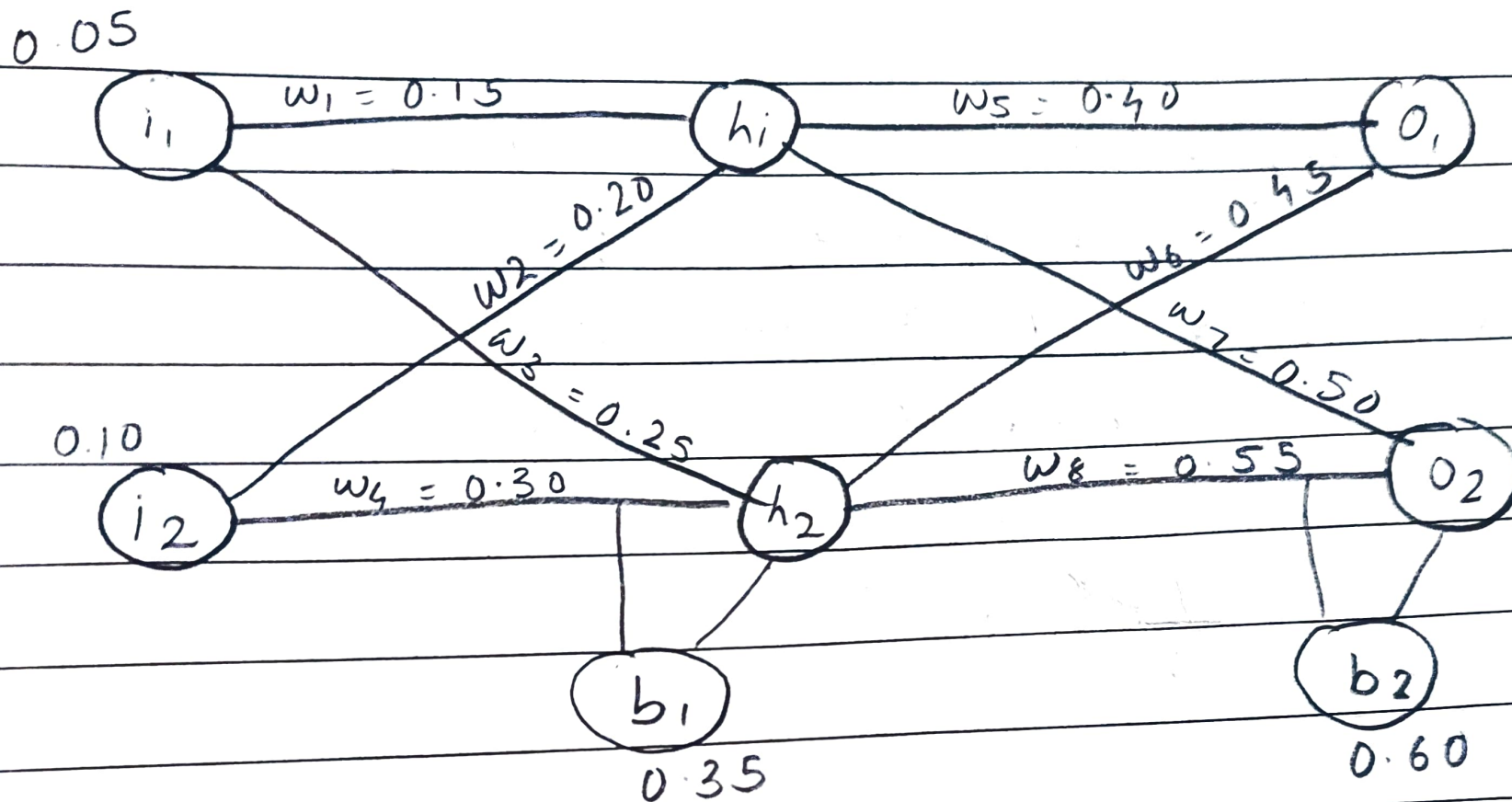




Target values : $T_1 = 0.01$

$T_2 = 0.99$

$\alpha = 0.5$

Forward propagation

$$h_1 = w_{11}i_1 + w_{21}i_2 + b_1 \\ = \underline{\underline{0.3775}}$$

$$h_2 = w_{31}i_1 + w_{41}i_2 + b_1 \\ = \underline{\underline{0.3925}}$$

$$\text{out}(h_1) = \frac{1}{1+e^{-h_1}} = \underline{\underline{0.5932}}$$

$$\text{out}(h_2) = \frac{1}{1+e^{-h_2}} = \underline{\underline{0.5968}}$$

$$o_1 = w_5 \text{out}(h_1) + w_6 \text{out}(h_2) + b_2 \\ = \underline{\underline{1.1058}}$$

$$o_2 = w_7 \text{out}(h_1) + w_8 \text{out}(h_2) + b_2 \\ = \underline{\underline{1.2248}}$$

$$\text{out}(o_1) = \frac{1}{1+e^{-o_1}} = \underline{\underline{0.7513}}$$

$$\text{out}(o_2) = \frac{1}{1+e^{-o_2}} = \underline{\underline{0.7729}}$$

$$E_{\text{total}} = \frac{1}{2} (t-p)^2$$

$$= \frac{1}{2} (T_1 - \text{out}(o_1))^2 + \frac{1}{2} (T_2 - \text{out}(o_2))^2$$

$$= \underline{\underline{0.2983}}$$

There is an error, so we do back propagation.

Backward propagation of outer layer

$$w_x^* = w_x - \alpha \left(\frac{\partial \text{Error}}{\partial w_x} \right)$$

$$w_5^* = w_5 - \alpha \left(\frac{\partial E_{\text{total}}}{\partial w_5} \right)$$

$$\frac{\partial E_{\text{total}}}{\partial w_5} = \frac{\partial E_{\text{total}}}{\partial \text{out } O_1} \times \frac{\partial \text{out } O_1}{\partial O_1} \times \frac{\partial O_1}{\partial w_5}$$

$$\frac{\partial E_{\text{total}}}{\partial \text{out } O_1} = - \left[\frac{\partial}{\partial \text{out}(O_1)} \left[\frac{1}{2} (\text{out}(O_1) - T_1)^2 + \frac{1}{2} (\text{out}(O_2) - T_2)^2 \right] \right]$$

$$= \text{out}(O_1) - T_1 = \underline{\underline{0.7413}}$$

$$\frac{\partial \text{out}(O_1)}{\partial O_1} = \frac{\partial}{\partial O_1} \left[\frac{1}{1 + e^{-O_1}} \right]$$

$$= \text{out}(O_1) [1 - \text{out}(O_1)] = \underline{\underline{0.1868}}$$

$$\frac{\partial O_1}{\partial w_5} = \frac{\partial}{\partial w_5} [w_5 \text{out}(h_1) + w_6 \text{out}(h_2) + b_2]$$

$$= \text{out}(h_1) = \underline{\underline{0.5932}}$$

$$\therefore \frac{\partial E_{\text{total}}}{\partial w_5} = 0.7413 \times 0.1868 \times 0.5932$$

$$= \underline{\underline{0.0821}}$$

$$w_5^* = 0.40 - 0.5 \times 0.0821$$

$$= \underline{\underline{0.3589}}$$

$$w_6^* = w_6 - \alpha \left(\frac{\partial E_{total}}{\partial w_6} \right)$$

$$\frac{\partial E_{total}}{\partial w_6} = \frac{\partial E_{total}}{\partial out O_1} \times \frac{\partial out O_1}{\partial O_1} \times \frac{\partial O_1}{\partial w_6}$$

$$\frac{\partial O_1}{\partial w_6} = \frac{\partial}{\partial w_6} [w_5 out(h_1) + w_6 out(h_2) + b_2]$$

$$= out(h_2) = \underline{\underline{0.5968}}$$

$$\begin{aligned} \frac{\partial E_{total}}{\partial w_6} &= 0.7413 \times 0.1868 \times 0.5968 \\ &= \underline{\underline{0.0826}} \end{aligned}$$

$$\begin{aligned} w_6^* &= 0.45 - 0.5 \times 0.826 \\ &= \underline{\underline{0.4087}} \end{aligned}$$

$$w_7^* = w_7 - \alpha \left(\frac{\partial E_{total}}{\partial w_7} \right)$$

$$\frac{\partial E_{total}}{\partial w_7} = \frac{\partial E_{total}}{\partial out O_2} \times \frac{\partial out O_2}{\partial O_2} + \frac{\partial O_2}{\partial w_7}$$

$$\frac{\partial E_{total}}{\partial out O_2} = out(O_2) - T_2 = \underline{\underline{+0.2171}}$$

$$\frac{\partial out(O_2)}{\partial O_2} = out(O_2) [1 - out(O_2)] = \underline{\underline{0.1755}}$$

$$\frac{\partial O_2}{\partial w_7} = \frac{\partial}{\partial w_7} [w_7 \text{Out}(h_1) + w_8 \text{Out}(h_2) + b_2]$$

$$= \text{Out}(h_1) = \underline{\underline{0.5932}}$$

$$\frac{\partial E_{\text{total}}}{\partial w_7} = -0.2171 \times 0.1755 \times 0.5932$$

$$= \underline{\underline{-0.0226}}$$

$$w_7^* = 0.50 - 0.5(-0.0226)$$

$$= \underline{\underline{0.5113}}$$

$$w_8^* = w_8 - \alpha \left(\frac{\partial E_{\text{total}}}{\partial w_8} \right)$$

$$\frac{\partial E_{\text{total}}}{\partial w_8} = \frac{\partial E_{\text{total}}}{\partial \text{out } O_2} \times \frac{\partial \text{out } O_2}{\partial O_2} \times \frac{\partial O_2}{\partial w_8}$$

$$\frac{\partial O_2}{\partial w_8} = \text{Out}(h_2) = \underline{\underline{0.5968}}$$

$$\frac{\partial E_{\text{total}}}{\partial w_8} = -0.2171 \times 0.1755 \times 0.5968$$

$$= \underline{\underline{-0.0227}}$$

$$w_8^* = 0.55 - 0.5 \times (-0.0227)$$

$$= \underline{\underline{0.5613}}$$

Back propagation of hidden layer.

$$\frac{\partial E_{total}}{\partial w_1} = \frac{\partial E_{total}}{\partial out_{h1}} \times \frac{\partial out_{h1}}{\partial h_1} \times \frac{\partial h_1}{\partial w_1}$$

$$\frac{\partial E_{total}}{\partial out_{h1}} = \frac{\partial E_{o1}}{\partial out(h_1)} + \frac{\partial E_{o2}}{\partial out(h_1)}$$

$$\frac{\partial E_{o1}}{\partial out(h_1)} = \frac{\partial E_{o1}}{\partial o_1} \times \frac{\partial o_1}{\partial out(h_1)}$$

$$= \frac{\partial E_{total}}{\partial out_{o1}} \times \frac{\partial out_{o1}}{\partial o_1} \times \frac{\partial o_1}{\partial out(h_1)}$$

$$\frac{\partial o_1}{\partial out(h_1)} = - \frac{\partial}{\partial out(h_1)} [w_5 out(h_1) + w_6 out(h_2) + b_2]$$

$$= w_5 = \underline{\underline{0.3589}}$$

$$\therefore \frac{\partial E_{o1}}{\partial out(h_1)} = 0.7513 \times 0.1868 \times 0.3589$$

$$= \underline{\underline{0.0503}}$$

$$\frac{\partial E_{o2}}{\partial out(h_1)} = \frac{\partial E_{total}}{\partial out_{o2}} \times \frac{\partial out_{o2}}{\partial o_2} \times \frac{\partial o_2}{\partial out(h_1)}$$

$$\frac{\partial o_2}{\partial out(h_1)} = - \frac{\partial}{\partial out(h_1)} [w_7 out(h_1) + w_8 out(h_2) + b_2]$$

$$= w_7 = \underline{\underline{0.5113}}$$

$$\frac{\partial E_{02}}{\partial \text{out}(h_1)} = (-0.2171) \times 0.1755 \times 0.5113$$

$$= -0.0194$$

$$\frac{\partial E_{\text{total}}}{\partial \text{out } h_1} = 0.0503 - 0.0194 = \underline{\underline{0.0309}}$$

$$\frac{\partial \text{out } h_1}{\partial h_1} = \frac{\partial}{\partial h_1} \left(\frac{1}{1 + e^{-h_1}} \right)$$

$$= \text{out}(h_1) [1 - \text{out}(h_1)] = \underline{\underline{0.2413}}$$

$$\frac{\partial h_1}{\partial w_1} = \frac{\partial}{\partial w_1} (w_1 i_1 + w_2 i_2 + b_1)$$

$$= i_1 = \underline{\underline{0.05}}$$

$$\frac{\partial E_{\text{total}}}{\partial w_1} = 0.0309 \times 0.2413 \times 0.05$$

$$= \underline{\underline{0.0004}}$$

$$w_1^* = w_1 - \alpha \left(\frac{\partial E_{\text{total}}}{\partial w_1} \right)$$

$$= 0.15 - 0.5 \times 0.0004$$

$$= \underline{\underline{0.1498}}$$

$$\frac{\partial E_{\text{total}}}{\partial w_2} = \frac{\partial E_{\text{total}}}{\partial \text{out } h_1} \times \frac{\partial \text{out } h_1}{\partial h_1} \times \frac{\partial h_1}{\partial w_2}$$

$$\frac{\partial h_1}{\partial w_2} = i_2 = \underline{\underline{0.1}}$$

$$\frac{\partial E_{total}}{\partial w_2} = 0.0309 \times 0.2413 \times 0.1$$

$$= \underline{\underline{0.0007}}$$

$$w_2^* = w_2 - 2 \left(\frac{\partial E_{total}}{\partial w_2} \right)$$

$$= 0.20 - 0.5 \times 0.0007$$

$$= \underline{\underline{0.19965}}$$

$$\frac{\partial E_{total}}{\partial w_3} = \frac{\partial E_{total}}{\partial out_{h2}} \times \frac{\partial out_{h2}}{\partial h_2} \times \frac{\partial h_2}{\partial w_3}$$

$$\frac{\partial E_{total}}{\partial out_{h2}} = \frac{\partial E_{o1}}{\partial out_{h2}} + \frac{\partial E_{o2}}{\partial out_{h2}}$$

$$\frac{\partial E_{o1}}{\partial out_{h2}} = \frac{\partial E_{total}}{\partial out_{o1}} \times \frac{\partial out_{o1}}{\partial o_1} \times \frac{\partial o_1}{\partial out(h_2)}$$

$$\frac{\partial o_1}{\partial out(h_2)} = w_6 = \underline{\underline{0.4087}}$$

$$\frac{\partial E_{o1}}{\partial out_{h2}} = 0.7513 \times 0.1868 \times 0.4087$$

$$= \underline{\underline{0.0573}}$$

$$\frac{\partial E_{o2}}{\partial out_{h2}} = \frac{\partial E_{total}}{\partial out_{o2}} \times \frac{\partial out_{o2}}{\partial o_2} \times \frac{\partial o_2}{\partial out(h_2)}$$

$$\frac{\partial O_2}{\partial \text{out}(h_2)} = w_8 = \underline{\underline{0.5613}}$$

$$\begin{aligned} \frac{\partial E_{o2}}{\partial \text{out} h_2} &= (-0.2171) \times 0.1755 \times 0.5613 \\ &= \underline{\underline{-0.0213}} \end{aligned}$$

$$\frac{\partial E_{\text{total}}}{\partial \text{out} h_2} = 0.0573 - 0.0213 = \underline{\underline{0.036}}$$

$$\frac{\partial \text{out}(h_2)}{\partial h_2} = \text{out}(h_2) [1 - \text{out}(h_2)] = \underline{\underline{0.2406}}$$

$$\frac{\partial h_2}{\partial w_3} = i_1 = \underline{\underline{0.05}}$$

$$\begin{aligned} \frac{\partial E_{\text{total}}}{\partial w_3} &= 0.036 \times 0.2406 \times 0.05 \\ &= \underline{\underline{0.0004}} \end{aligned}$$

$$\begin{aligned} w_3^* &= w_3 - 2 \left(\frac{\partial E_{\text{total}}}{\partial w_3} \right) \\ &= 0.25 - 0.5 \times 0.0004 \\ &= \underline{\underline{0.2498}} \end{aligned}$$

$$\frac{\partial E_{\text{total}}}{\partial w_4} = \frac{\partial E_{\text{total}}}{\partial \text{out} h_2} \times \frac{\partial \text{out} h_2}{\partial h_2} \times \frac{\partial h_2}{\partial w_4}$$

$$\frac{\partial h_2}{\partial w_4} = i_2 = 0.1$$

$$\begin{aligned}\frac{\partial E_{total}}{\partial w_4} &= 0.036 \times 0.2406 \times 0.1 \\ &= \underline{\underline{0.0008}}\end{aligned}$$

$$\begin{aligned}w_4^* &= w_4 - 2 \left(\frac{\partial E_{total}}{\partial w_4} \right) \\ &= 0.30 - 0.5 \times 0.0008 \\ &= \underline{\underline{0.2996}}\end{aligned}$$